

LED BLINK

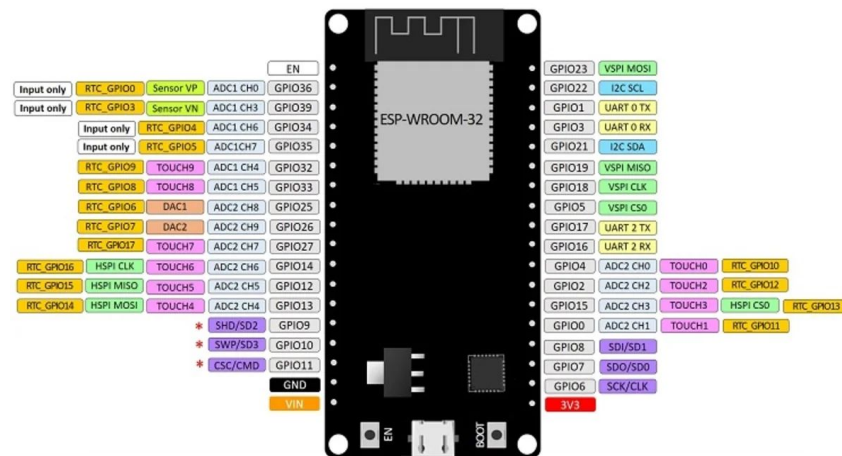
LED Blinking example

ESP32 GPIO Pins

- Total GPIO pins exposed: 36
- Usable pins for general applications: ~30
- Digital I/O capable pins: 24
- Input-only pins: 6
 - These are: **GPIO34, GPIO35, GPIO36, GPIO37, GPIO38, and GPIO39**
 - They **cannot output** a voltage (e.g., for driving an LED).

Suitable for output (e.g., controlling devices like LEDs):

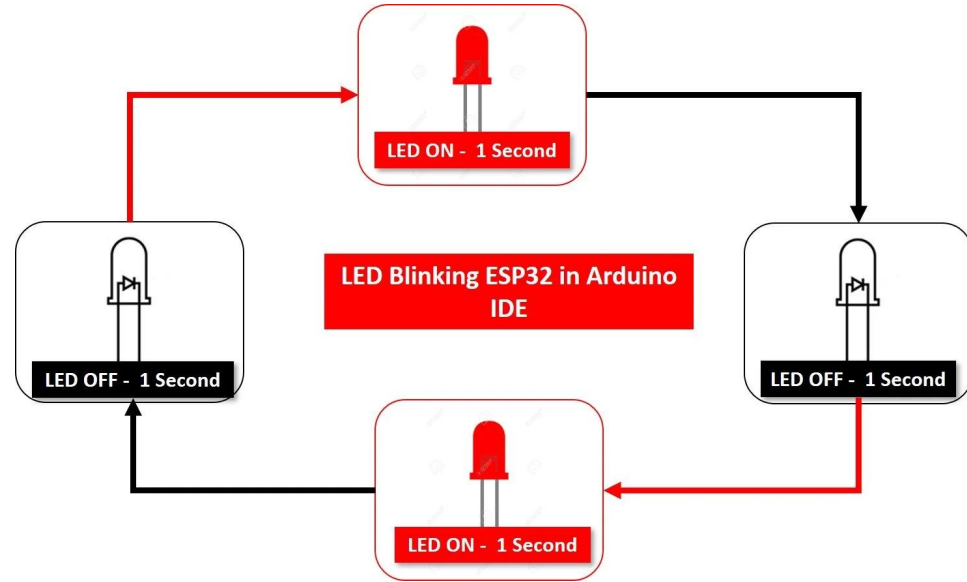
- The remaining **24 GPIOs** (excluding the 6 above) can function as both **digital input and output**, making them versatile for most embedded applications.



LED Blinking Example Using ESP32

In this example, we will control an LED with GPIO22 of ESP32. That means we will learn to use GPIO pins as digital output pins. We connect an LED to GPIO22 through a current limiting resistor of 220 ohms. This ESP32 LED blinking example works as mentioned below:

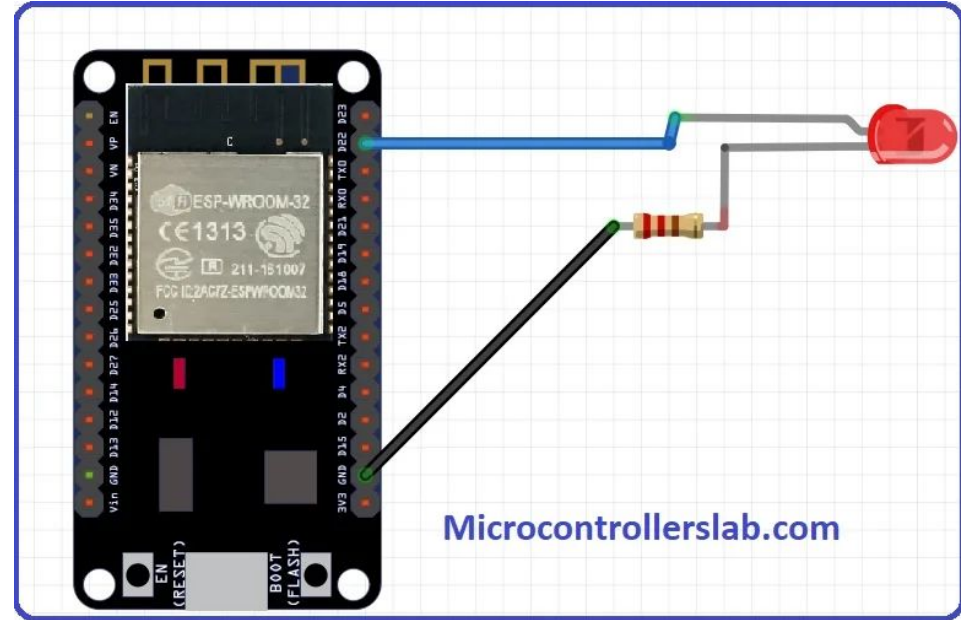
- RED LED Turns on for one second which means GPIO22 goes to an active high state
- In the next step, the LED turns off for one second which means GPIO22 goes to an active low state
- The above two steps keep repeating as shown below:



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```
void setup()
{
  pinMode(22, OUTPUT); // Set GPIO22 as digital output pin
}
```

```
void loop()
{
  digitalWrite(22, HIGH); // Set GPIO22 active high
  delay(1000); // delay of one second
  digitalWrite(22, LOW); // Set GPIO22 active low
  delay(1000); // delay of one second
}
```

GAS SENSOR

OVERVIEW OF MQ2 GAS SENSOR

The MQ2 gas sensor is a useful device that can detect the levels of various gases, including LPG, smoke, alcohol, propane, hydrogen, methane, and carbon monoxide, in the surrounding environment. It offers two output options: a digital output pin and an analog output pin.

It's important to note that the MQ2 gas sensor doesn't provide separate information about each gas. Instead, it gives us a general indication of the gas combination or the presence of gases as a whole.

By using the MQ2 sensor, we can determine if there is a gas leak or if the air quality is poor. This information is valuable as it allows us to take appropriate measures to ensure our safety, such as activating an alarm or turning on ventilation systems.



PINOUT

The MQ2 gas sensor consists of four pins that serve different functions:

- ◆ **VCC pin:** It should be connected to the power supply's positive terminal (5V) to provide the necessary power to the sensor.
- ◆ **GND pin:** It should be connected to the power supply's negative terminal (0V) to complete the electrical circuit.
- ◆ **DO pin:** This is a digital output pin that indicates the presence of flammable gases. When the gas concentration is detected, the output is set to **LOW**, and when there is no gas detected, the output is set to **HIGH**. The threshold for gas concentration detection can be adjusted using a potentiometer that is built into the sensor.
- ◆ **AO pin:** This is an analog output pin that generates a voltage signal proportional to the gas concentration. When the gas concentration increases, the voltage output also increases, and when the gas concentration decreases, the voltage output decreases accordingly.





Additionally, the MQ2 gas sensor includes two LED indicators:

- ◆ PWR-LED indicator: This LED serves as a power indicator, indicating whether the sensor is receiving power. When the sensor is properly powered, the PWR-LED is turned on, providing visual confirmation of the sensor's operational state.
- ◆ DO-LED indicator: This LED is directly linked to the DO pin of the sensor. It indicates the presence of gas concentration based on the value received from the DO pin. When the gas concentration is detected and the DO pin is set to **LOW**, the DO-LED turns on. Conversely, when there is no gas concentration detected and the DO pin is set to **HIGH**, the DO-LED turns off.

HOW IT WORKS

Regarding the DO pin:

- ◆ The MQ2 module includes a built-in potentiometer for adjusting the sensitivity or threshold of gas concentration.
- ◆ When the gas concentration in the surrounding environment exceeds the set threshold, the output pin of the sensor becomes **LOW**, and the DO-LED turns on.
- ◆ Conversely, when the gas concentration falls below the threshold, the output pin becomes **HIGH**, and the DO-LED turns off.

Regarding the AO pin:

- ◆ The voltage on the AO pin increases as the gas concentration rises.
- ◆ Conversely, as the gas concentration decreases, the voltage on the AO pin decreases accordingly.

It's important to note that the potentiometer adjustment does not affect the value on the AO pin.

The MQ2 Sensor Warm-up

The MQ2 gas sensor requires a warm-up period before it can be used effectively. Here are the details:

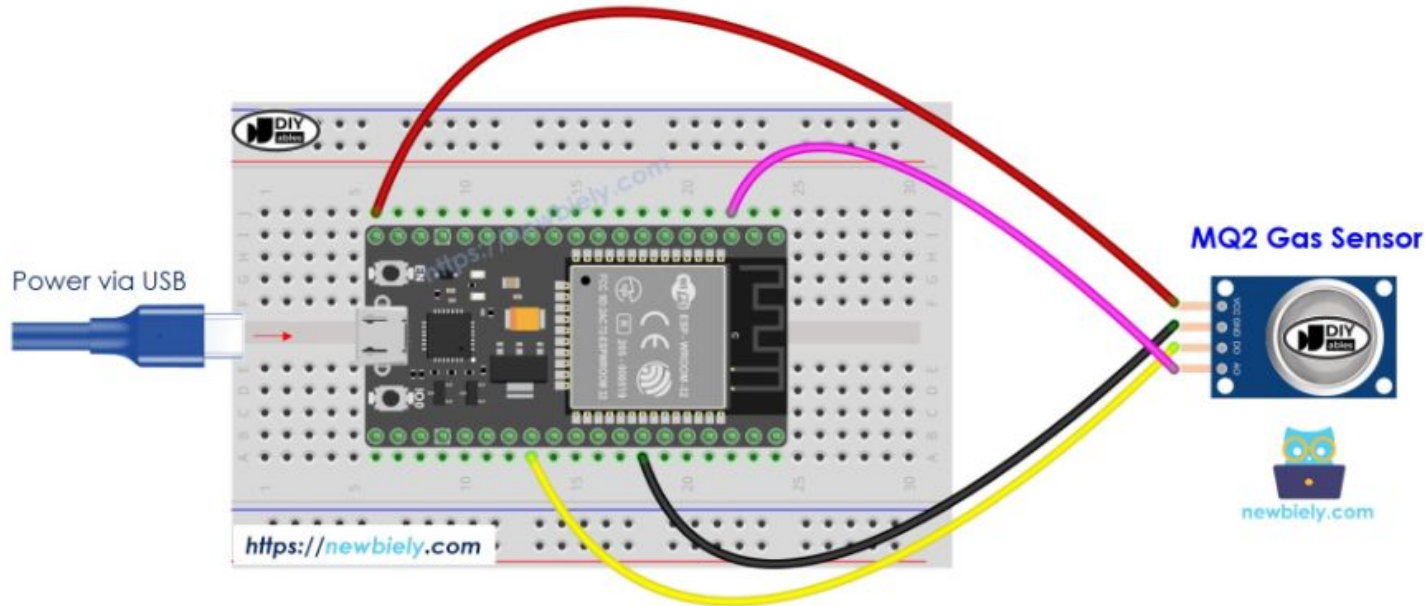
- ◆ When using the sensor for the first time after a long period of storage (around a month or more), it is necessary to warm it up for 24-48 hours. This extended warm-up time ensures accurate operation.
- ◆ If the sensor has been used recently, the warm-up time is significantly shorter. It typically takes only 5-10 minutes for the sensor to fully warm up. During this warm-up period, the sensor may initially provide high readings, but these readings will gradually decrease until the sensor stabilizes.

To warm up the MQ2 sensor, simply connect its **VCC** and GND pins to a power supply or connect them to the **VCC** and GND pins of an ESP32. Allow the sensor to remain in this state for the required warm-up period.

Wiring Diagram

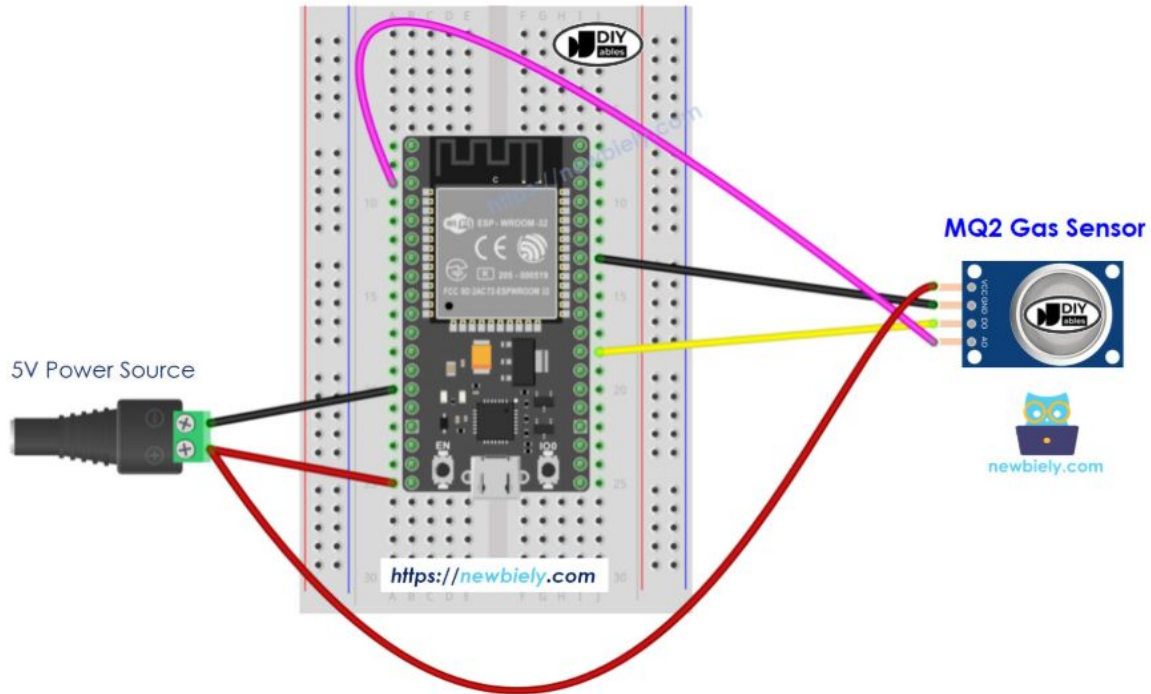
Since the MQ2 gas sensor module has two outputs, you can choose to use one or both of them, depending on what you need.

◆ How to connect ESP32 and gas sensor using [breadboard](#) (powered via USB cable)



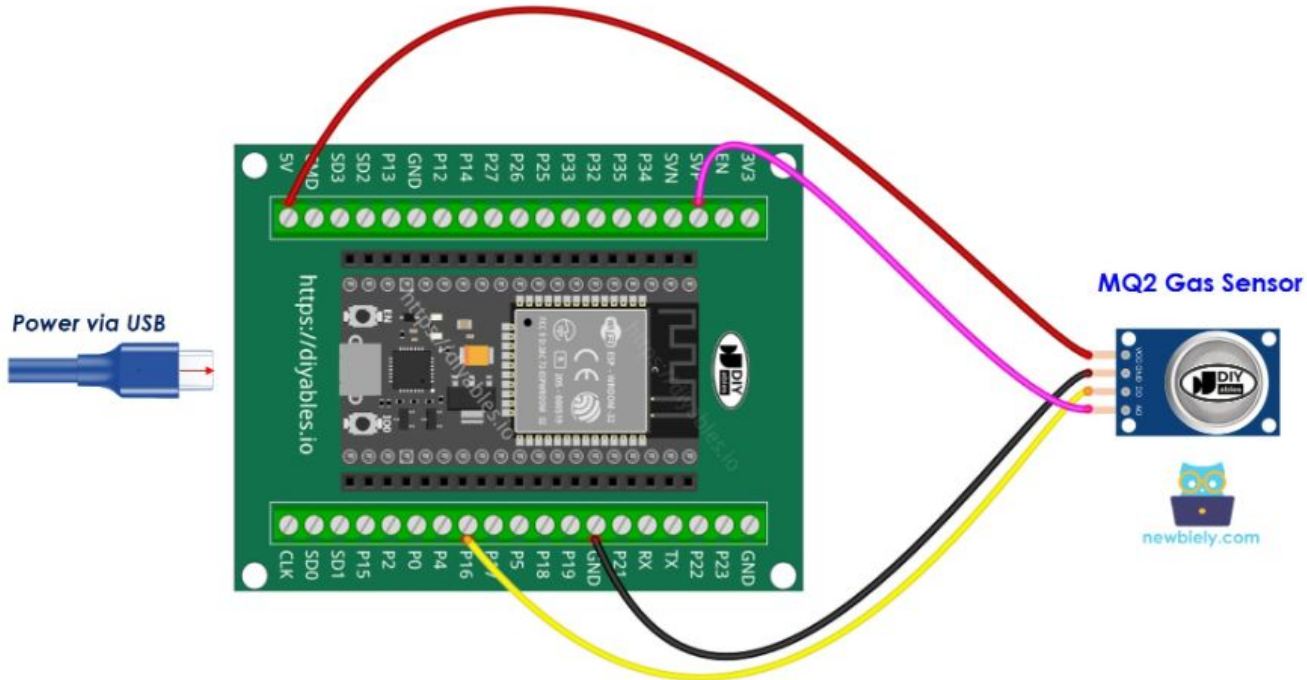
Wiring Diagram

How to connect ESP32 and gas sensor using breadboard (powered via Vin pin)



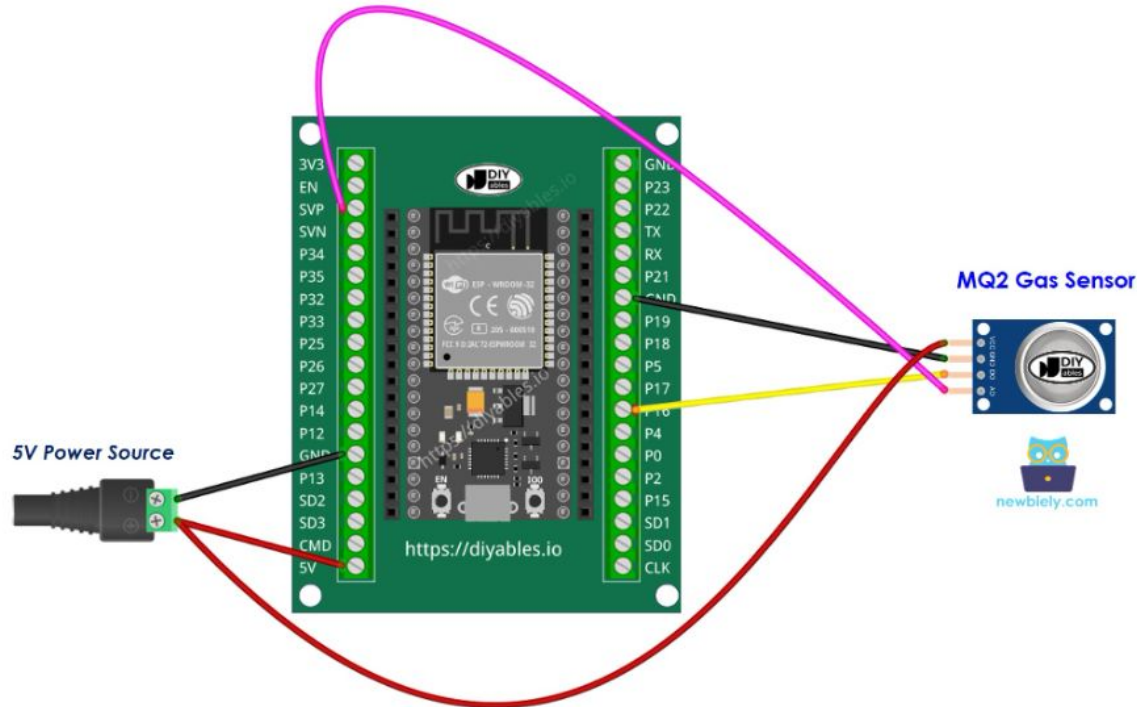
Wiring Diagram

- How to connect ESP32 and gas sensor using [screw terminal block breakout board](#) (powered via USB cable)



Wiring Diagram

How to connect ESP32 and gas sensor using [screw terminal block breakout board](#) (powered via Vin pin)



Code : Reading value through DO only

```
#define DO_PIN 16 // ESP32's pin GPIO16 connected to DO pin of the MQ2 sensor
```

```
void setup() {  
  // initialize serial communication  
  Serial.begin(9600);  
  // initialize the ESP32's pin as an input  
  pinMode(DO_PIN, INPUT);  
  
  Serial.println("Warming up the MQ2 sensor");  
  delay(20000); // wait for the MQ2 to warm up  
}
```

```
void loop() {  
  int gasState = digitalRead(DO_PIN);  
  
  if (gasState == HIGH)  
    Serial.println("The gas is NOT present");  
  else  
    Serial.println("The gas is present");  
}
```



COM6



Send

The gas is NOT present
The gas is NOT present
The gas is NOT present
The gas is NOT present
The gas is NOT present
The gas is present
The gas is present
The gas is present
The gas is present
The gas is present

Code: Reading value through A0 only

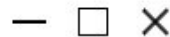
```
#define AO_PIN 36 // ESP32's pin GPIO36 connected to AO pin of the MQ2 sensor
```

```
void setup() {  
  // initialize serial communication  
  Serial.begin(9600);  
  
  // set the ADC attenuation to 11 dB (up to ~3.3V input)  
  analogSetAttenuation(ADC_11db);  
  Serial.println("Warming up the MQ2 sensor");  
  delay(20000); // wait for the MQ2 to warm up  
}
```

```
void loop() {  
  int gasValue = analogRead(AO_PIN);  
  
  Serial.print("MQ2 sensor AO value: ");  
  Serial.println(gasValue);  
}
```



COM6



Send

```
MQ2 sensor A0 value: 135
MQ2 sensor A0 value: 136
MQ2 sensor A0 value: 136
MQ2 sensor A0 value: 573
MQ2 sensor A0 value: 674
MQ2 sensor A0 value: 938
MQ2 sensor A0 value: 954
MQ2 sensor A0 value: 1000
MQ2 sensor A0 value: 1002
MQ2 sensor A0 value: 1014
MQ2 sensor A0 value: 1017
```



Autoscroll



Show timestamp

Newline



9600 baud



Clear output

